

Utility-Scale Solar PV Subcontractor Management: Comprehensive Simulation Data for Project HELIOS

1. Subcontractor Operations Overview

The execution of a utility-scale solar photovoltaic (PV) engineering, procurement, and construction (EPC) project operates on an industrial assembly-line model distributed across hundreds of hectares. On a 100 MW site such as Project HELIOS in Uzbekistan, the main EPC contractor functions as the central integrator and commercial manager, while the physical execution is entirely delegated to specialized subcontractors. Understanding the operational psychology, workflow, and output generation of these entities is critical for accurately simulating their behavior.

1.1 Organizational Deployment and Field Roles

A subcontractor operating on a 100 MW solar site deploys a distinct organizational hierarchy tailored to highly repetitive, geographically spread tasks. The typical structure includes a Site Manager who acts as the primary interface with the main EPC contractor, driving commercial claims, schedule adherence, and resource mobilization. Beneath the Site Manager, discipline-specific Site Engineers translate the Issued for Construction (IFC) drawings into actionable daily work fronts.

The operational backbone consists of Foremen and Supervisors who manage gangs of laborers or specific machinery fleets. Quality Assurance/Quality Control (QA/QC) Engineers are deployed to ensure the work meets technical specifications and to compile handover documentation, such as pour cards or torque records. Health, Safety, and Environment (HSE) Officers monitor compliance with site safety rules, which is particularly critical in regions subject to extreme weather and heavy machinery operations¹. The labor force itself is divided into skilled trades (e.g., excavator operators, certified high-voltage jointers, topographic surveyors) and unskilled labor responsible for manual material distribution.

1.2 Work Order Scope versus Daily Operations

The legal boundary of a subcontractor's involvement is defined by the Work Order (WO) or Subcontract Agreement. The WO dictates the macro-parameters: the total Bill of Quantities (BOQ), the baseline schedule, commercial unit rates, Liquidated Damages (LDs) provisions, and technical specifications³.

Conversely, daily operations represent the micro-execution of the WO within the chaotic reality of a construction site. While the WO might dictate the installation of 35,000 single-axis tracker piles over three months, daily operations are dictated by available "work fronts" (areas cleared by preceding trades), material availability, and localized constraints. Subcontractors are inherently cash-flow driven; their daily operational strategy often involves "cherry-picking" the easiest, highest-yielding BOQ items to maximize the monthly Interim Payment Certificate (IPC) rather than strictly following the logical critical path⁴.

1.3 Internal Workflow versus Main Contractor Supervision

The internal workflow of a subcontractor is strictly production-oriented. The primary objective is to achieve maximum daily output to generate maximum billing under the BOQ. This involves morning toolbox talks, allocating plant and labor to specific blocks, executing the work at high velocity, and conducting internal snagging to prepare for formal inspection.

The main EPC contractor's workflow, however, is assurance-oriented. It focuses on validating that the subcontractor's output meets the quality baseline and aligns with the master schedule. The main contractor issues Permits to Work (PTW), conducts joint inspections, issues Non-Conformance Reports (NCRs) for deviations, and certifies completed quantities for payment⁴. This fundamental dichotomy—production speed versus quality assurance—is the primary source of friction between subcontractors and main contractors.

1.4 Daily Reporting Obligations

Every subcontractor is contractually obligated to submit a Daily Progress Report (DPR). The DPR serves as the atomic unit of project tracking and the foundational document for future commercial claims. It chronicles the quantitative output achieved (e.g., piles driven, trenches excavated), the resources deployed (manpower headcount, equipment utilized versus idle), materials consumed or received, and health and safety incidents. Crucially, subcontractors use the DPR to record blockers or constraints caused by the main contractor or other trades, which forms the documentary basis for Extension of Time (EOT) claims⁵.

2. Civil Subcontractor (GROUNDWORK)

The civil subcontractor establishes the physical foundation of the site. In the context of Project HELIOS, GROUNDWORK operates across an expansive area, typically requiring the grading and preparation of 200 to 250 hectares for a 100 MW array, governed by Uzbekistan's KMK 3.01.01-97 construction norms⁶.

2.1 Exhaustive Scope of Work

The scope for GROUNDWORK encompasses all earthmoving, drainage, and concrete activities outside the main 220kV substation footprint.

Activity Category	Specific Tasks Executed
Site Clearance & Grubbing	Removal of vegetation, boulders, and topsoil stripping across the entire array footprint to facilitate pile driving.
Bulk Earthworks	Cut and fill operations to achieve design gradients. Single-axis trackers generally require a north-south slope of less than 10-15% to prevent torque tube binding ⁸ .

Internal Roads & Hardstandings	Construction of perimeter security roads, internal access roads, and inverter station pads, including sub-grade preparation, sub-base compaction, and gravel wearing courses.
Drainage Systems	Excavation of swales, installation of culverts at road crossings, and rip-rap placement to manage storm-water runoff and prevent soil erosion around foundations.
Inverter Station Foundations	Excavation, blinding, rebar fixing, formwork erection, and concrete pouring for the skid-mounted inverter and medium-voltage transformer pads.
Perimeter Fencing	Topographical alignment, post hole excavation, concrete footing casting, and erection of high-security fencing and gates.
Control Room Building	Traditional civil works including shallow foundations, masonry or steel superstructure, roofing, plumbing, and architectural finishing.

2.2 Civil Work Sequence

Civil execution follows a strict sequence from macro-landscaping to micro-foundations. The sequence begins with topographical surveying and boundary pegging. Clearing and grubbing follow, removing obstacles that would otherwise impede pile drivers. Bulk earthworks are then executed to establish the required topographical profiles.

Simultaneously, primary drainage channels are excavated to prevent waterlogging during the construction phase, and the primary access road network is established to allow heavy material deliveries (e.g., modules and steel) to reach the array blocks. Once the macro-grading is complete, localized excavation for inverter station foundations occurs, allowing concrete to cure early in the project schedule. Perimeter fencing is erected to secure the site, and finally, the graded array blocks are formally handed over to the mechanical subcontractor (EREKTOR).

2.3 Daily Output and Productivity Rates

The daily output of a civil subcontractor is highly dependent on the deployment of heavy machinery. Assuming a fully mobilized fleet (e.g., 10 excavators, 4 motor graders, 4 vibratory compactors, 2 mobile batching plants, and over 100 personnel), the expected outputs are substantial.

Civil Activity	Realistic Daily Output	Justification / Context
Site Clearing	3.0 – 5.0 Hectares	Depends on vegetation density. Scrubland in Uzbekistan allows rapid clearing.
Internal Roads	150 – 250 Linear Meters	Includes sub-base laying, watering to optimal moisture content (OMC), and vibratory compaction.
Drainage Trenching	300 – 400 Linear Meters	Achieved using 20-ton excavators equipped with trapezoidal buckets.
Concrete Pouring	40 – 60 m^3	Lower than mass concrete outputs due to the highly distributed nature of small inverter pads requiring transit mixer travel time across the 250-hectare site.
Fencing	100 – 150 Linear Meters	Includes post-hole digging, post setting, concrete pouring, and mesh stretching.

2.4 Document Production

GROUNDWORK produces a massive volume of QA/QC documentation daily. Beyond the DPR, the subcontractor generates Requests for Inspection (RFIs) to summon the main contractor's CHECKER to approve sub-grade compaction, rebar placement, and formwork prior to concrete pours. Concrete Pour Cards are generated for every foundation, detailing slump test results, ambient temperatures, and total volumes. Survey Reports, generated by RTK GPS rovers, are submitted to verify that the final coordinates and elevations of the earthworks match the IFC drawings.

2.5 Measurement and Billing Structure

Civil works are billed based on the remeasurement of actual executed quantities against the

BOQ. Earthworks (cut and fill) are measured in cubic meters (m^3) via topographical survey cross-sections comparing pre-construction and post-construction elevations. Roads and fences are measured in linear meters (m). Concrete is measured in m^3 based strictly on the theoretical dimensions provided in the IFC drawings; subcontractors are not paid for "over-break" (excess concrete poured due to excavating holes too wide).

2.6 Typical Causes of Delay

The most common delay for GROUNDWORK involves geotechnical surprises. Encountering subterranean hard rock or cobble layers that were not identified in the initial geotechnical report necessitates a shift from rapid excavation to slow pre-drilling or hydraulic breaking⁸. Weather is another primary constraint. Rain immediately halts compaction activities because the soil exceeds its Optimal Moisture Content (OMC), rendering compaction impossible. Conversely, in the arid summer conditions of Uzbekistan, extreme heat triggers mandatory work suspensions between 11:00 and 16:00 per state labor regulations, compressing the operational window¹.

3. MMS / Module Erection Subcontractor (EREKTOR)

EREKTOR is responsible for the Module Mounting Structure (MMS) and the photovoltaic modules. On Project HELIOS, this involves horizontal single-axis trackers (HSAT), which demand significantly higher spatial precision than fixed-tilt systems to ensure the continuous torque tube rotates without binding¹¹.

3.1 Scope of Work

EREKTOR's scope is highly repetitive but requires meticulous logistical coordination.

- **Pile Distribution and Driving:** Logistics of moving heavy steel piles from laydown yards to array locations and driving them to design refusal and embedment depths using hydraulic machines¹³.
- **Static Load Testing:** Conducting pull-out, lateral, and compression tests on a representative sample of driven piles to verify geotechnical holding capacity against wind uplift forces¹⁴.
- **Tracker Assembly:** Installing bearing adapter seats, placing torque tubes, splicing tube sections, and mounting the central slew drives and electric motors¹¹.
- **Module Mounting:** Unpacking pallets, distributing panels along the rows, and clamping PV modules to the mounting purlins or rails.
- **Torqueing and Alignment:** Ensuring the tracker rotates freely across its full axis and applying specified torque to all structural bolts, followed by the application of paint witness marks for QA verification¹⁷.

3.2 Single-Axis Tracker Erection Sequence

1. **Survey Staking:** Topographers peg exact pile locations. The tolerance is exceptionally strict (typically ± 6 mm for pile head elevation and ± 20 mm east-west) to prevent structural binding of the torque tube¹⁷.
2. **Pile Driving:** Self-propelled hydraulic pile drivers (e.g., Gayk, Turchi models producing

950-1200 Joules of impact energy) drive the galvanized steel C or H profiles²⁰.

3. **Bearing Housing Installation:** Articulating bearing assemblies are attached to the pile heads to support the rotating mechanism¹¹.
4. **Torque Tube Placement:** Heavy torque tube sections are lifted via telehandlers, dropped into the bearings, and spliced together.
5. **Drive Unit Installation:** The slew drive and motor are mounted at the center of the row, anchoring the drive assembly and transferring rotational force to the torque tube¹¹.
6. **Module Installation:** Modules are fixed using specific mid-clamps and end-clamps.
7. **QA & Torqueing:** A final check of the rotation arc is performed. Calibrated "click" torque wrenches apply the designated torque (e.g., 14-20 Nm for module clamps)¹⁸.

3.3 Daily Output and Productivity Rates

Productivity depends heavily on the mechanization of the fleet and the hardness of the soil. A typical 100-person crew supported by heavy machinery yields the following:

Erection Activity	Realistic Daily Output	Justification / Context
Pile Driving (Sandy/Clay Soil)	300 – 450 Piles	Utilizing 3-4 hydraulic pile drivers (e.g., Turchi 300F). A single machine can drive 100-150 piles/day in good conditions ²⁰ .
Pile Driving (Rocky Soil)	90 – 180 Piles	Output plummets if Down-The-Hole (DTH) pre-drilling is required to bypass refusal layers ⁸ .
Tracker Table Assembly	15 – 25 Rows	A mechanical crew can assemble the torque tubes and bearings for roughly 1.5 MW per day.
Module Installation	2,500 – 3,500 Modules	Equates to roughly 1.5 to 2.1 MW per day (assuming modern 600W+ high-power modules) ²⁰ .

3.4 Document Production and Measurement

EREKTOR produces Pile Driving Logs daily, recording the coordinates, final depth achieved, and hydraulic pressure or blow counts for every single pile. Torque Records provide QA sign-offs confirming that bolts were tensioned correctly and marked. MMS billing is highly discrete and

easily verified. Subcontractors are paid per unit: per pile driven (numbers), per tracker structure fully assembled (numbers or tonnes of steel), and per PV module installed (numbers or MW capacity).

3.5 Resource Plan

A highly mobile equipment fleet is required. The resource plan includes 3 to 4 GPS-guided hydraulic pile drivers, 2 to 3 all-terrain telehandlers for lifting torque tubes, tractors with flatbed trailers for distributing module pallets, and dozens of calibrated torque wrenches. The crew comprises machine operators, material handlers (unskilled labor), mechanical assemblers, module installers, and QA inspectors.

3.6 Typical Causes of Delay

Pile refusal is the most catastrophic delay for EREKTOR. Hitting underground cobbles or bedrock necessitates a switch from rapid driving to slow, expensive pre-drilling, disrupting the entire assembly line²³. Wind is another critical constraint; crane lifts and the handling of large-format PV modules must cease when wind speeds exceed 10 m/s due to the dangerous "sail effect"¹¹. Logistical bottlenecks, such as delayed module deliveries or broken pallets, quickly halt the fast-moving installation crews.

4. Electrical Erection Subcontractor (ELECTRA)

ELECTRA bridges the gap between solar generation and the grid, handling DC strings, inverters, and the medium voltage (MV) alternating current (AC) collection networks. Their work requires certified technicians and carries the highest risk of catastrophic failure if quality is compromised.

4.1 Exhaustive Scope of Work

- **DC Cabling:** Pulling 4 mm^2 or 6 mm^2 cross-linked polyethylene (XLPE) solar cables from the module strings to Combiner Boxes or directly to string inverters²⁵.
- **Cable Management:** Installing galvanized wire mesh cable trays or clipping cables directly to the tracker torque tube using UV-resistant ties.
- **DC Terminations:** Crimping MC4 connectors onto the DC cables using highly calibrated tools to ensure low contact resistance and prevent arc faults²⁶.
- **MV Cable Laying (33kV):** Excavating trenches (typically 80-100 cm deep per local norms), laying sand bedding, pulling armored MV XLPE cables, laying protective bricks or warning tape, and backfilling²⁸.
- **MV Terminations and Jointing:** Executing highly sensitive cold-shrink or heat-shrink straight-through joints in the trenches and end terminations at the equipment³⁰.
- **Substation E&M Erection:** Placing, aligning, and bolting skid-mounted inverters, 33/220kV step-up transformers, Gas Insulated Switchgear (GIS) or Air Insulated Switchgear (AIS), control panels, and relay panels within the substation building.
- **Testing and Commissioning:** Performing Megger testing (insulation resistance), continuity checks, open-circuit voltage (V_{oc}) measurements, and IV curve tracing in accordance with IEC 62446-1 standards³².

4.2 Electrical Installation Sequence

The sequence operates simultaneously above and below ground. Below ground, the process

follows: trench excavation → 150 mm sand bedding → pulling armored MV cables using winches and rollers → placing warning tape/tiles → backfilling and compaction³⁴. Above ground, cable trays are installed along the tracker rows. DC cables are pulled from the modules along the trays to the inverter locations. Certified electricians then crimp the MC4 connectors and plug them into the modules.

In parallel, heavy equipment such as inverters and AC collection panels are craned into position. Once the cables are laid, specialized jointers perform the MV cold-shrink jointing in the

trenches and the end terminations at the equipment³⁰. Finally, commissioning tests (IR, V_{oc} , polarity) are executed before any energization occurs³².

4.3 Daily Output and Productivity Rates

Electrical productivity is split between bulk pulling and precision jointing.

Electrical Activity	Realistic Daily Output	Justification / Context
DC Cable Pulling	3,000 – 5,000 Meters	High velocity achieved by manual pulling teams feeding off drum jacks along the tracker rows.
MV Cable Pulling (33kV)	600 – 1,000 Meters	Highly dependent on the trench being fully prepared by GROUNDWORK. Requires mechanical winches and straight-line rollers to manage Sidewall Bearing Pressure (SWBP) ²⁸ .
MC4 Terminations	200 – 300 Connectors	Performed by trained electricians using calibrated crimping tools ²⁷ .
MV Cable Jointing	2 – 4 Joints per Jointer	MV cold-shrink jointing is meticulous. It requires extreme cleanliness, precise stripping of the semi-conductive layer, and careful application of

		stress-control mastics ³⁰ .
--	--	--

4.4 Measurement and Billing Structure

Electrical billing is structured linearly for cables (measured per linear meter pulled) and volumetrically for trenches (measured in m^3 of excavation and backfill). Terminations, straight-through joints, combiner boxes, and inverter installations are billed per discrete unit (number).

4.5 Typical Causes of Delay

Trench clashes are a major source of delay. ELECTRA cannot pull continuous lengths of stiff MV cable if GROUNDWORK has not completed road crossings or drainage culverts. Cable drum logistics also cause significant delays; 33kV cable drums are massive and require specialized hydraulic jacks and trailers. If the civil subcontractor leaves the access roads in poor condition, moving these drums becomes impossible³⁷. Furthermore, MV jointing requires highly certified technicians. A shortage of these specialists creates severe bottlenecks, as uncertified personnel cannot legally or safely perform the work.

5. Substation Civil Subcontractor (GRIDCON)

The 33/220kV Grid Substation (GSS) is a distinct, highly concentrated work front. Unlike the sprawling solar array, the GSS requires high-precision civil engineering, governed by strict norms such as Uzbekistan's KMK regulations³⁸.

5.1 Scope of Work

- **Equipment Foundations:** Casting high-precision reinforced concrete pads for the 220kV main step-up transformers (including fire blast walls and oil catch pits), switchgear, circuit breakers, and steel transmission gantries.
- **Earthing Grid (Earthmat):** Deep excavation and laying of the primary copper or copper-clad steel earth grid to achieve an overall earth resistance of $< 1\Omega$ as per IEEE Std 80 and IEC standards⁴⁰.
- **Cable Trenches:** Construction of cast-in-situ reinforced concrete trenches with removable checkered plate covers to route control and power cables safely.
- **Substation Control Building:** Full turnkey construction of the control room, battery room, and relay building.
- **Perimeter and Finishing:** Erecting high-security perimeter walls, heavy-duty access gates, and spreading switchyard gravel (yard stone) to suppress step-and-touch potentials.

5.2 Substation Civil Sequence

The GSS sequence begins with the mass grading and compaction of the substation footprint. Because the Earthmat underpins the entire facility, the deep trenches for the copper grid must be excavated and the conductors laid and exothermically welded before other foundations block access⁴¹. Once the earthmat is tested and backfilled, deep foundations for transmission gantries and the massive main transformers are poured. Shallow foundations for isolators and

breakers follow. The control building superstructure is erected concurrently. Finally, the interconnected concrete cable trenches are cast, and the yard gravel is spread before demobilization.

5.3 Document Production and Billing

GRIDCON produces standard civil documentation (pour cards, concrete cube compressive strength test results, rebar approvals). Critically, they must produce Earth Resistance Test

records, utilizing the Fall-of-Potential method to prove the earthmat achieves the $< 1\Omega$

target⁴¹. Billing follows standard civil remeasurement principles: m^3 of concrete, kilograms of rebar steel, and square meters of formwork and masonry.

6. Subcontractor–Main Contractor Interface

The interface between the EPC main contractor and the subcontractors acts as the operational gearbox of the project; friction here severely degrades the project's velocity.

6.1 Work Front Handover Process

A solar project operates on a sequential assembly-line model. GROUNDWORK grades a block and hands it to EREKTOR for piling, who then hands it to ELECTRA for wiring. This transfer is governed by a formal "Work Front Handover Certificate." The preceding subcontractor clears the area of debris and machinery. The succeeding subcontractor inspects the site alongside the main contractor's QA/QC team. If the area meets the IFC tolerances (e.g., sub-grade is flat, no windrows block the pile lines), the certificate is signed. If minor defects exist, they are noted on a punch list for the preceding subcontractor to rectify within a specified timeframe.

6.2 Daily Interface Meetings

At 16:00 or 17:00 daily, the Main Contractor's Project Manager hosts a mandatory coordination meeting. The agenda involves reviewing the day's DPRs against the baseline targets, resolving spatial clashes (e.g., EREKTOR's telehandlers blocking ELECTRA's cable pulling path), and planning the logistics for the next 48 hours.

6.3 Billing Claim Review (JMR and CHECKER Process)

Subcontractors submit weekly or monthly billing claims. The process begins with a Joint Measurement Record (JMR). The subcontractor's Quantity Surveyor (QS) and the main contractor's QS (the CHECKER) physically walk the site to agree on the executed quantities (e.g., counting installed tracker tables, measuring trench lengths using a measuring wheel)⁴. Once the JMR is signed, it is attached to the Interim Payment Certificate (IPC) application. The EPC Project Manager and Commercial Manager review the claim. They deduct the contractual retention amount (usually 5-10%), apply any back-charges (e.g., if the main contractor had to supply diesel to the subcontractor), and certify the net amount for payment.

6.4 Method Statement Deviations and NCRs

If a subcontractor deviates from the approved Method Statement—for example, if EREKTOR begins pre-drilling holes wider than specified without using the required concrete grout—the main contractor's QA team issues a Non-Conformance Report (NCR). Work on that specific activity is immediately halted. The subcontractor must submit a corrective action plan, which

must be approved by the Engineer of Record, before work can resume.

6.5 Performance Monitoring and Liquidated Damages (LDs)

The main contractor tracks progress via S-curves, plotting planned cumulative progress against actual cumulative progress. If a subcontractor's actual progress falls significantly (typically 10-15%) below the baseline, the EPC issues a formal notice demanding a "Recovery Programme."

If the subcontractor fails to recover and delays the critical path, Liquidated Damages (LDs) are levied. Under standard subcontracts (often mirroring FIDIC Sub-Clause 8.7), LDs apply a financial penalty of 0.5% to 1% of the total subcontract value per week of delay, capped at a maximum of 10%⁴⁴. LDs are typically enforced only if the subcontractor's delay directly threatens the main contractor's handover date to the client (e.g., the off-taker NEGU in Uzbekistan)⁴⁵.

7. Subcontractor Document Production — Full Detail

7.1 Daily Progress Report (DPR)

The DPR contains:

- **Activities Done:** Specific blocks or zones worked on.
- **Quantities:** Exact output metrics measured against daily targets.
- **Resources Deployed:** Manpower broken down by trade (foremen, skilled, unskilled) and Equipment broken down by operational status (active, idle, breakdown).
- **Issues:** Weather delays, material shortages, or pending RFIs.

7.2 Weekly Billing Claim (IPC) Format

The billing claim is a financial spreadsheet containing:

- **Header:** Project Name, Date, IPC Number, Subcontractor Name.
- **Body:** BOQ Item Number, Description of Work, Unit of Measurement (UoM), BOQ Quantity, and Contractual BOQ Rate.
- **Progress Metrics:** Quantity Certified Previous, Quantity Claimed This Week, Quantity Cumulative.
- **Financials:** Amount Claimed This Week, Cumulative Amount, Less Retention, Less Advance Recovery, Net Amount Payable⁴.

7.3 Method Statement

A comprehensive engineering document detailing *how* a task will be safely executed:

- **Scope & Purpose.**
- **Resource Requirements:** Specific plant, equipment, and personnel required.
- **Methodology:** Step-by-step sequential instructions (e.g., from unloading MV cable drums to final backfilling).
- **Inspection Test Plan (ITP):** Defining the mandatory QA/QC hold points where the main contractor must witness the work.

7.4 Hazard Identification and Risk Assessment (HIRA)

A safety document that identifies task-specific hazards (e.g., heavy lifting, trench collapse, electrocution). It assesses the risk severity and likelihood, and dictates mandatory mitigation

measures (e.g., trench shoring, insulated tools, lockout/tagout procedures).

7.5 Material and Labour Records

- **Material Delivery Record:** Logs Goods Received Notes (GRNs), verifying the quantities of modules, steel, or cables against shipping manifests and identifying transit damage.
- **Labour Deployment Record:** Timesheets detailing hours worked per individual. These are used to calculate productivity (man-hours per unit) and verify compliance with state labor laws.

8. Productivity & Resource Planning

8.1 Baseline Productivity Rates

Activity	Realistic Productivity Rate	Key Variables & Justifications
Pile Driving (Driven)	150 – 200 piles / day	Assumes sandy/alluvial soil using modern hydraulic drivers (e.g., Turchi/Gayk) ²⁰ .
Pile Driving (Pre-Drill)	30 – 60 piles / day	Required in hard rock. Down-the-Hole (DTH) drilling severely limits output ⁸ .
Concrete (Inverter Pads)	40 – 50 m^3 / day	Hindered by the logistics of moving transit mixers between distributed pad locations across 200 hectares.
Tracker Erection	20 tables / day	Assumes a dedicated 40-man mechanical assembly crew ²⁰ .
Module Installation	1.5 – 2.0 MW / day	Equates to ~2,500–3,500 modules installed by a highly synchronized 40-man crew ²⁰ .
DC Cable Pulling	3,000 – 5,000 m / day	Highly manual process; cables are light and easily pulled along trays.

MV Cable Pulling	600 – 1,000 m / day	Requires mechanical winches. Rate drops if trenches are incomplete or bending radii are tight ²⁸ .
MV Cable Termination	2 – 4 joints / day	Per certified joiner. Cold-shrink technology improves speed, but careful layer stripping remains time-intensive ³⁰ .

8.2 Weather Impacts on Productivity

- **Extreme Heat:** Under Uzbekistan's SanPiN No. 0289-10 and State Labour Inspectorate mandates, outdoor work is heavily restricted or entirely suspended between 11:00 and 16:00 when ambient temperatures exceed 36°C¹. This forces subcontractors to adopt split shifts (early morning and late evening), reducing overall daily efficiency by 15-20% due to lost momentum.
- **Wind:** Gusts exceeding 10 m/s suspend module handling due to the "sail effect," which poses severe safety risks to laborers. Crane operations for erecting transformers or inverters are also halted.
- **Rain:** Precipitation immediately halts earthworks compaction (altering the OMC) and open-trench MV cable jointing due to the risk of moisture contamination inside the splice⁴⁷.

8.3 Recovery Programmes

When a subcontractor falls behind schedule, the main contractor formally demands a recovery program. Development involves three primary tactics:

1. **Crashing:** Mobilizing additional plant and labor. This is highly effective but financially penalizes the subcontractor.
2. **Fast-tracking:** Executing sequential tasks in parallel (e.g., pulling cables in half-completed trenches while GROUNDWORK finishes the other half). This increases logistical friction.
3. **Shift Work:** Introducing night shifts under artificial lighting. While this increases available working hours, productivity per hour is significantly lower, and safety risks escalate.

9. Simulating Realistic Subcontractor Behaviour

To construct authentic AI agents simulating construction subcontractors, the simulation must incorporate real-world tactics, defenses, and commercial survival behaviors. Subcontractors optimize for cash flow and limit their liability.

9.1 Realistic Excuses for Delays

Subcontractors rarely admit fault; they shift blame to external factors or the main contractor.

- **GROUNDWORK:** *"State Labour Inspectorate regulations prohibit work in 38°C heat. We had to stand down the compaction crews at noon to comply with SanPiN 0289-10."*¹.

- **EREKTOR:** *"The work front was not handed over cleanly. GROUNDWORK left windrows blocking our pile drivers, and the surveyor pegs are missing."*
- **EREKTOR:** *"We hit refusal at 1.2 meters in Block 4. The geotechnical report indicated alluvial sand, but we are hitting solid cobble. We need an approved variation order to commence DTH pre-drilling."⁸*
- **ELECTRA:** *"We cannot pull the 33kV cables today; the main contractor's CHECKER has not signed off on the trench bed compaction RFI."*

9.2 Billing Inflation Tactics

Subcontractors attempt to maximize their monthly IPCs to maintain cash flow.

- **Front-loading:** Overstating the completion percentage on early, high-value BOQ items. For example, ELECTRA might claim 100% completion for MV trenching when the warning tape and final backfill are still incomplete.
- **Geotechnical Reclassification:** GROUNDWORK frequently attempts to classify standard compacted soil as "hard rock" to trigger the significantly higher BOQ excavation rates.
- **Wastage Deflection:** If ELECTRA runs out of DC cable due to poor cutting optimization on site, they will claim the manufacturer "short-shipped" the drums to avoid paying for the wasted material.

9.3 Quality Shortcuts

When pushed by aggressive schedules, subcontractors take covert shortcuts.

- **Skipping Torque Procedures:** EREKTOR crews may use pneumatic impact drivers to blast module clamps tight instead of using calibrated torque wrenches. This saves time but risks crushing the aluminum frame lip (voiding the module warranty) or failing to achieve the required 14-20 Nm preload¹⁸.
- **Trench Bedding Omissions:** ELECTRA might lay MV cables directly on rocky trench bottoms instead of placing the required 150 mm sand bedding. This saves sand and time but virtually guarantees a future earth fault when the rocks puncture the XLPE insulation under backfill pressure²⁸.
- **Shallow Piling (The "Cut-Off"):** If EREKTOR's pile driver hits refusal just 30 cm short of the design embedment, the crew might covertly cut the top of the pile off with a grinder rather than extracting it and pre-drilling. This leaves the foundation vulnerable to wind uplift failure¹³.

9.4 Legitimate Grievances

Subcontractors often have valid complaints against the main contractor.

- **IFC Revisions:** The EPC's engineering team continually revising drawings (e.g., changing string layouts) *after* the subcontractor has already executed the work, causing uncompensated rework.
- **Free-Issue Material Delays:** The main contractor failing to deliver modules, inverters, or steel on time, leaving EREKTOR and ELECTRA paying daily wages to an idle workforce.
- **Payment Withholding:** The EPC withholding a massive IPC payment over trivial, unrelated punch-list items, starving the subcontractor of the cash flow needed to pay

their laborers.

9.5 Dispute Escalation and Resource Triggers

Disputes escalate linearly. A subcontractor begins by issuing formal letters citing specific subcontract clauses (e.g., referencing unforeseeable physical conditions under FIDIC)⁴⁸. If ignored, the subcontractor will submit a formal claim for an Extension of Time (EOT) with associated prolongation costs. The ultimate escalation is a "go-slow" strike or completely demobilizing from the site pending adjudication⁵.

Subcontractors almost never mobilize additional resources voluntarily due to the massive capital outlay required. Mobilization is triggered only by:

1. A formal, legally vetted threat from the EPC to apply Liquidated Damages.
2. The clearing of overdue IPC payments, injecting fresh cash into the subcontractor's accounts.
3. The approval of a highly lucrative Variation Order (VO) that guarantees profit on the extra resources.

10. Real Document Templates — Full Detail

The following markdown templates represent the exact administrative formats used on utility-scale solar projects. These can be utilized directly by AI agents simulating administrative workflows.

10.1 Daily Progress Report (DPR) Templates

GROUNDWORK - DPR Format

DAILY PROGRESS REPORT - GROUNDWORK

Date: [YYYY-MM-DD] | **Block:** [Block ID] | **Weather:** [Temp, Conditions]

1. Resources Deployed

- **Manpower:** [X] Supervisors, [Y] Heavy Equipment Operators, [Z] General Labourers
- **Equipment:** [X] Excavators (Active/Idle), [Y] Motor Graders (Active/Idle), [Z] Compactors

2. Production Metrics

Activity	UoM	Daily Target	Achieved Today	Cumulative
Topsoil Stripping & Grubbing	Ha	4.0	3.5	45.5
Internal Roads (Sub-base)	m	200	180	1,200

MV Trench Excavation	m	350	300	2,100
Inverter Pad Concrete	m ³	45	45	360

3. Issues & Constraints

- Excavator #3 breakdown (blown hydraulic hose at 10:00).
- Work suspended 11:30-16:00 due to extreme heat (38°C) per SanPiN 0289-10 labor law.
- Pending RFI approval from EPC CHECKER for Block 4B sub-grade compaction.

EREKTOR - DPR Format

DAILY PROGRESS REPORT - EREKTOR

Date: [YYYY-MM-DD] | **Block:** [Block ID] | **Weather:** [Temp, Wind Speed]

1. Resources Deployed

- **Manpower:** [X] Foremen, [Y] Mechanical Assemblers, [Z] Module Installers
- **Equipment:** [X] Hydraulic Pile Drivers, [Y] Telehandlers, [Z] Tractors

2. Production Metrics

Activity	UoM	Daily Target	Achieved Today	Cumulative
Piles Driven (Target Depth)	Nos	400	280	4,500
Torque Tubes Assembled	Rows	20	22	210
Slew Drives Installed	Nos	20	20	200
PV Modules Installed	Nos	3000	3150	25,000

3. Issues & Constraints

- Pile driving slowed significantly in Block B4 due to cobble refusal; 15 piles require DTH

- pre-drilling.
- Wind gusts exceeded 11 m/s at 14:00; all module lifting and handling suspended for 2 hours for safety.
- Shortage of M8 mid-clamps supplied by EPC; module installation gang shifted to Block C1.

ELECTRA - DPR Format

DAILY PROGRESS REPORT - ELECTRA

Date: [YYYY-MM-DD] | Block: [Block ID] | Weather: [Temp, Conditions]

1. Resources Deployed

- **Manpower:** [X] Certified Electricians, [Y] MV Jointers, [Z] Cable Pullers

2. Production Metrics

Activity	UoM	Daily Target	Achieved Today	Cumulative
DC Cable Pulling (4mm ²)	m	4000	4200	35,000
MV Cable Laying (33kV)	m	800	650	4,200
MC4 Terminations Crimped	Nos	250	260	2,100
MV Cold Shrink Joints	Nos	4	3	12
String Voc Testing	Str	50	50	450

3. Issues & Constraints

- Waiting on trench clearance from GROUNDWORK at Road Crossing #4; cannot pull continuous MV run.
- Shortage of 6mm² DC cable drums in the main laydown area.
- Dust storm contaminated trench at Joint Pit 3; jointer spent 2 hours cleaning area before applying stress mastic.

DAILY PROGRESS REPORT - GRIDCON

Date: [YYYY-MM-DD] | **Area:** [220kV GSS] | **Weather:** [Temp, Conditions]

1. Resources Deployed

- **Manpower:** [X] Steel Fixers, [Y] Formwork Carpenters, [Z] Concrete Gang

2. Production Metrics

Activity	UoM	Daily Target	Achieved Today	Cumulative
Earthmat Trench/Laying	m	100	120	800
Transformer Pad Rebar	Kg	2000	2100	15,000
Cable Trench Concrete	m ³	30	30	150
Perimeter Wall Masonry	m ²	50	45	400

3. Issues & Constraints

- Rebar delivery delayed by 4 hours; steel fixers diverted to formwork prep for the GIS building.
- Earth resistance testing delayed; waiting for EPC engineer to witness Fall-of-Potential test.

10.2 Weekly Billing Claim (IPC) Format

INTERIM PAYMENT CLAIM (IPC)

Subcontractor: [Subcontractor Name] | **Claim No:** 04 **Period:** [YYYY-MM-DD] to [YYYY-MM-DD] | **Project:** HELIOS 100MW

BOQ Ref	Description of Work	UoM	BOQ Rate (\$)	Prev. Qty	Qty This Wk	Cumul. Qty	Amount This Wk (\$)

2.1.3	Supply and Drive HSAT Piles	Nos	12.50	4,220	1,450	5,670	18,125.00
2.2.1	Assemble Tracker Torque Tubes	Rows	85.00	188	110	298	9,350.00
2.3.4	Install 600W PV Module	Nos	3.20	21,850	9,500	31,350	30,400.00

Subtotal Claimed This Week: \$57,875.00 Less 5% Contractual Retention: (\$2,893.75) Less EPC Diesel Back-charges: (\$450.00) Net Amount Payable: \$54,531.25

Attachments: Joint Measurement Records (JMR) signed and stamped by EPC Quantity Surveyor.

10.3 Work Front Handover Certificate

WORK FRONT HANDOVER CERTIFICATE

Certificate No: HC-045 | **Date:** [YYYY-MM-DD] **From (Handing Over):** GROUNDWORK **To (Receiving):** EREKTOR **Location/Block:** Array Block 4B

QA/QC Checklist:

- [x] Area graded to IFC design elevations (Tolerance $\pm 50\text{mm}$).
- [x] Topsoil and windrows entirely cleared from pile lines.
- [x] Access roads compacted to allow telehandler movement without bogging.
- [x] Boundary pegs and control points established by surveyor.

Punch List / Outstanding Minor Items:

1. Small soil stockpile to be removed from Northeast corner by [Date]. Does not impede pile driving.

Signatures: Groundwork Rep: _____ Erektor Rep: _____ EPC QA/QC: _____

10.4 Subcontractor Performance Report

MONTHLY SUBCONTRACTOR

PERFORMANCE REPORT

Subcontractor: ELECTRA | **Month:** [Month, Year] | **Evaluator:** EPC Project Manager

1. Schedule & Progress (Rating: 3/5)

- **Planned Cumulative:** 45% | **Actual Cumulative:** 41%
- **Comments:** Fell behind on MV trenching due to mechanical winch breakdowns. However, DC cable pulling is slightly ahead of the baseline schedule.

2. Quality & Workmanship (Rating: 4/5)

- **NCRs Issued:** 1 (Minor issue with MC4 crimping tool calibration log missing).
- **Comments:** MV cold-shrink jointing is of excellent quality. Megger testing pass rate is 99.5%.

3. Health, Safety & Environment (Rating: 2/5)

- **Incidents:** 2 First-Aid cases (Heat exhaustion).
- **Comments:** Failed to strictly enforce the 11:00-16:00 heat suspension protocol on two occasions, violating SanPiN 0289-10. Formal written warning issued.

4. Commercial & Administration (Rating: 3/5)

- **Comments:** JMRs are submitted on time, but claims often aggressively inflate variations for standard cable routing adjustments around tracker drive motors.

Overall Action Plan: ELECTRA must submit a formal schedule recovery plan for MV cable laying within 3 working days. The Safety Officer is required to conduct a mandatory heat-stress and hydration toolbox talk for all pulling gangs.

Works cited

1. Uzbekistan Bans Mass Outdoor Work During Extreme Heat, <https://www.uzdaily.uz/en/uzbekistan-bans-mass-outdoor-work-during-extreme-heat/>
2. Labor code of the Republic of Uzbekistan - CIS Legislation, <https://cis-legislation.com/document.fwx?rgn=145006>
3. EPC versus FIDIC Contracts: Key Differences - Sora & Associates, <https://www.sora.ro/2026/05/31/epc-versus-fidic-contracts/>
4. What Is An Interim Payment Certificate | Global Law Experts, <https://globallawexperts.com/what-is-an-interim-payment-certificate/>
5. (PDF) Resolving Disputes Under FIDIC Contracts: A Practical and, https://www.researchgate.net/publication/405291729_Resolving_Disputes_Under_FIDIC_Contracts_A_Practical_and_Procedural_Guide
6. KMK 3.02.01-97 | PDF - Scribd, <https://ru.scribd.com/document/625669760/%D0%9A%D0%9C%D0%9A-3-02-01-97>
7. WHK 3.02.01-19, https://mc.uz/uploads/mcuz_8223227347559.pdf
8. Ground for Solar: Geotechnical Assessment Guide | TURCHI,

- <https://turchi.pr.it/en/news/solar-farm-geotechnical-survey-guide>
9. Employers in Uzbekistan recommended to reduce outdoor work, <https://www.gazeta.uz/en/2026/07/16/labor/>
 10. Occupational Heat Stress Rules — Uzbekistan, <https://heatrules.com/countries/uz/>
 11. Horizontal Single - Axis Electrical Solar Tracker M - Metaloumin, <https://metaloumin.gr/wp-content/uploads/2026/09/Horizontal-Single-Axis-Electrical-Solar-Tracker.pdf>
 12. Single-Axis Tracker Installation Guide for Utility-Scale Solar Systems, <https://pvrrack.com/installation-guide/tracker-installation/>
 13. Solar Foundation Installation: Pile, Screw & Concrete - PVRack, <https://pvrrack.com/installation-guide/foundation-installation/>
 14. Pullout Test Method Statement for Solar Project | PDF - Scribd, <https://www.scribd.com/document/822300631/Allied-POT-Work-Method-Statement-for-20mW>
 15. PoT (Pull-out-test) - GeoTec Insitu, <https://geotec.rs/en/usluge/pot/>
 16. Pile Load Testing Methodology | PDF | Deep Foundation - Scribd, <https://www.scribd.com/document/438389792/Pull-Out-Test-Methodology>
 17. Plant Tracker Installation Guide | PDF | Nut (Hardware) - Scribd, <https://www.scribd.com/document/829823594/1-Installation-Manual-for-Tracker-System-1P-En-KST00618-V1-1-1>
 18. Solar Racking Fastener Torque Specs (2026): Bolt Grades ... - PVRack, <https://pvrrack.com/installation-guide/fastener-torque-specs/>
 19. Single Axis Solar Tracker System: How It Works & What to Buy, <https://solarvisionai.com/single-axis-solar-tracker-system/>
 20. Solar Farm Construction 2026: Cost per MW, Timeline, ROI - Turchi, <https://turchi.pr.it/en/solar-farm-construction>
 21. Turchi Pile Drivers: Solar & Guardrail Piling Equipment, <https://www.aleanda-europe.com/product-category/turchi-pile-drivers-machine/>
 22. Gayk pile drivers from Germany - Machineryline, <https://machineryline.uk/-/pile-drivers/Gayk/Germany--c92tm4908cntDE>
 23. Sinking solar - DNV, <https://www.dnv.com/article/sinking-solar-230080/>
 24. Case Study: Screw Piles vs Driven Posts on Rocky Sites - Anern Store, <https://www.anernstore.com/blogs/diy-solar-guides/screw-piles-driven-posts-rocky>
 25. Solar PV systems - DC cable sizing with examples - ELEK Software, <https://elek.com/articles/solar-pv-systems-dc-cable-sizing-calculations/>
 26. Solar Safety: Torque & MC4 Crimp Tool Guide - Anern Store, <https://www.anernstore.com/blogs/off-grid-solar-solutions/torque-mc4-crimp-tools-safety>
 27. Comprehensive Guide to Solar DC Connectors: Types and Uses, <https://www.handaconnector.com/solar-dc-connector-guide/>
 28. MV Cable Installation: SEC Standard Guide | ANPU, <https://apscable.com/news/mv-cable-installation-sec-standards/>
 29. The Effect of the Vertical Layout on Underground Cable Current, <https://www.mdpi.com/1996-1073/17/3/674>

30. How to Install MV Cable Termination Kits with Cold-Shrinkable, <https://www.haivofusecutout.com/How-to-Install-MV-Cable-Termination-Kits-with-Cold-Shrinkable-Accessories-id42263475.html>
31. Cable Jointing Step-by-Step Instructions for Nexans JTS, <https://etechcomponents.com/step-by-step-jointing-instructions-for-nexans-jts-heatshrinkable-mv-cable-joints/>
32. Solar PV Commissioning: IEC 62446 Tests Before Handover - MIMAH, <https://mimah.net/posts/solar-pv-commissioning-testing>
33. Solar PV System Commissioning Checklist: Ensuring Proper, <https://www.sunpal-energy.com/solar-pv-system-commissioning-checklist-ensuring-proper-installation-before-final-payment/>
34. Underground Cable Laying Method Statement, <https://electricalengineering123.com/underground-cable-laying-method-statement/>
35. Cable Laying - Methvin Production Rates, <https://methvin.co/production-rates/mechanical-electrical/cable/cable-laying/>
36. Termination Kit Installation Procedure for MV Cables, <https://axis-india.com/termination-kit-installation-medium-voltage-cables/>
37. MV Cable Laying Method for Solar Plant | PDF | Duct (Flow) - Scribd, <https://www.scribd.com/document/881701032/KOM-PVB-FQE-MS-MVAC-002-0-Method-of-statement-for-MV-AC-Cable-Trenching-and-Laying>
38. Uzbekistan Construction Regulations Guide | PDF | Concrete - Scribd, <https://www.scribd.com/document/976470063/Uzbekistan-Standards>
39. KMK 2.09.03-02, https://mc.uz/uploads/mcuz_49418148203342.pdf
40. Earthing (grounding) system according to IEC, BS-EN and IEEE, <https://cbm-technology.com.pl/en/good-earthing-grounding-technical-information/contractor-zone/earthing-grounding-system-according-to-iec-bs-en-and-ieee-standards/>
41. Earthing Resistance Standards for Substations | IEEE/IEC Guide, <https://industrialmonitordirect.com/blogs/knowledgebase/acceptable-earth-resistance-values-for-industrial-substations>
42. Solar Power Plant Earthing Test Report | PDF - Scribd, <https://www.scribd.com/document/968208513/Solar-Plant-Earthing-Test-Report>
43. Joint Measurement Report for Solar Project | PDF - Scribd, <https://www.scribd.com/document/704170165/JMR-Ph-3>
44. FIDIC Delay Damages Clause Explained | PDF - Scribd, <https://www.scribd.com/doc/190416168/Clause-8-7-Delay-Damages-Understanding-Clauses-in-FIDIC-Conditions-of-Contract-for-EPC-Turnkey-Projects-First-Edition-1999>
45. Sazagan Solar 1 - ACWA Power, <https://acwapower.com/en/what-we-do/projects/sazagan-solar-1/>
46. Effective Solar Farm Piling Best Practices for Optimized Performance, <https://everstarmachinery.com/blog/effective-solar-farm-piling-best-practices-for-optimized-performance/>
47. Cable Joints for Renewable Energy | Wind, Solar, BESS & MV Grid,

<https://www.lovink-enertech.com/en/applications/joints-for-renewable-energy/>
48. CLAUSE 20, DISPUTE RESOLUTION - FIDIC,
<https://fidic.org/sites/default/files/24%20CLAUSE%2020,%20DISPUTE%20RESOLUTION.pdf>